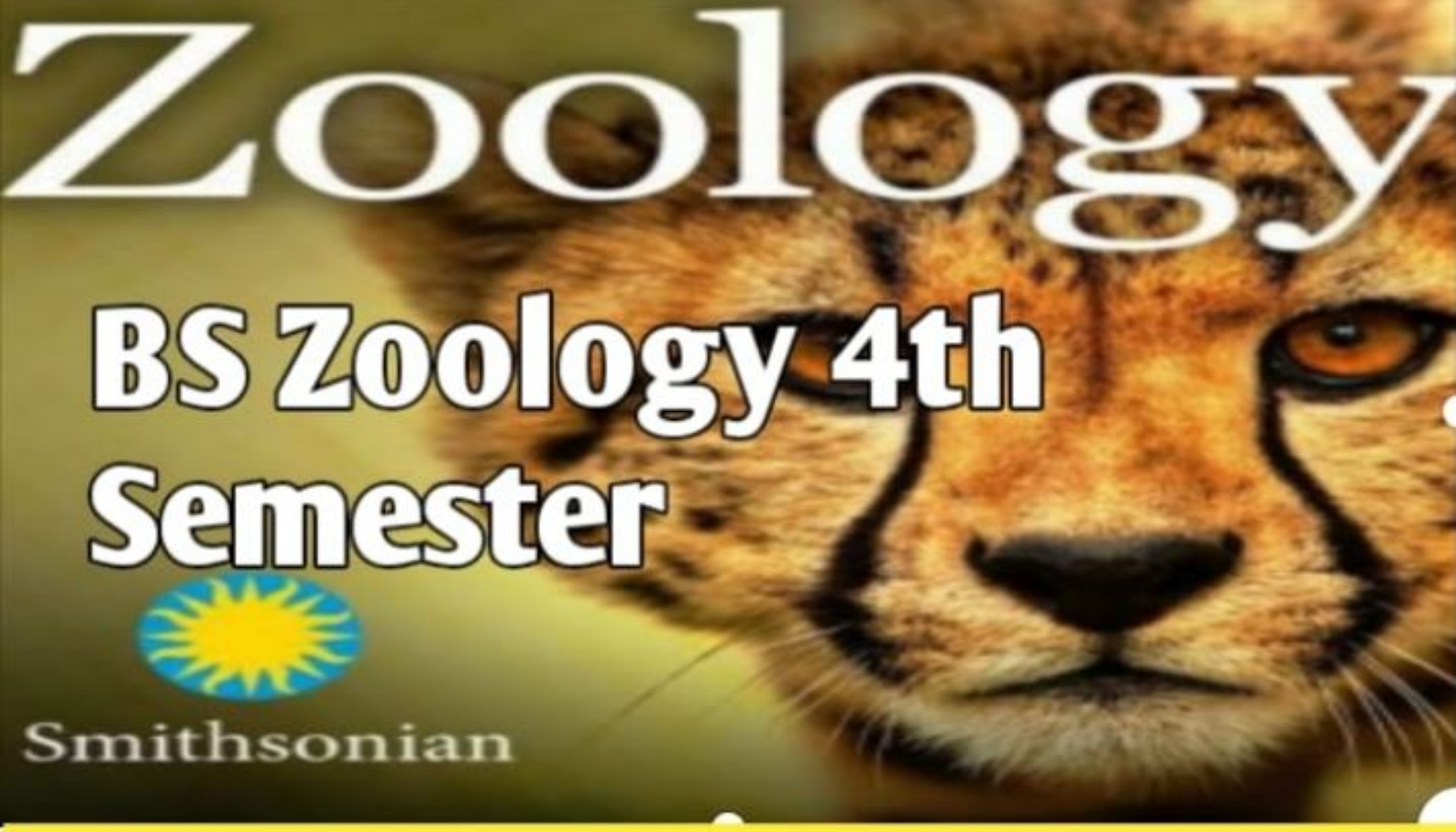


Prepared & Composed by Muhammad Habib

Contact At: 0307 8576495

The key to success is to focus our conscious mind on things we desire not things we fear.

BC200202403: AATIA SHARIF

ZOO202:Quiz no. 3

Question # 1 of 10 (Start time: 02:28:37 PM, 03 March 2022)

_____ directs sound to the middle ear.

Select the correct option

- ○ Ear drum
- ○ Inner ear
- ○ Cochlea
- ○ Tympanum

Type here to search

BC200202403: AATIA SHARIF

ZOO202:Quiz no. 3

Question # 2 of 10 (Start time: 02:29:38 PM, 03 March 2022)

The respiratory rate of a hibernating ground squirrel falls from 100 to 200 breaths per minute to about _____ breaths per minute.

Select the correct option

- 8
- 2
- 6
- 4

Type here to search

Question # 3 of 10 (Start time: 02:30:54 PM, 03 March 2022)

Mammals, like all amniotes, have a _____ kidney.

Select the correct option

- Complex
- Protonephric
- Simple
- Metanephric

Type here to search

ZOO202:Quiz no. 3

Question # 4 of 10 (Start time: 02:31:51 PM, 03 March 2022)

Some mammals had a large space, called the _____ separating the anterior and posterior teeth.

Select the correct option

- ○ Diastema
- ○ Cusps
- ○ Cynodonts
- ○ Cutting edges

Type here to search

BC200202403: AATIA SHARIF

ZOO202:Quiz no. 3

Question # 5 of 10 (Start time: 02:32:23 PM, 03 March 2022)

The sensory patch of the inner ear that contains the sound receptors is long and coiled and is called the

Select the correct option

- ○ Tympanum
- ○ Cochlea
- ○ Ear drum
- ○ Inner ear

Type here to search

ZOO202:Quiz no. 3

Question # 6 of 10 (Start time: 02:33:31 PM, 03 March 2022)

Beavers produce urine that is _____ as concentrated as blood

Select the correct option

○	Twice
○	Four times
○	Three times
○	Not

Type here to search

BC200202403: AATIA SHARIF

ZOO202:Quiz no. 3

Question # 7 of 10 (Start time: 02:34:28 PM, 03 March 2022)

Production through cellular mechanism is called _____thermogenesis.

Select the correct option

- Metabolic
- Non-shivering
- Shivering
- Muscular

Type here to search

ZOO202:Quiz no. 3

Question # 8 of 10 (Start time: 02:35:45 PM, 03 March 2022)

Which one of the following is true hibernator?

Select the correct option

- Human
- Monkey
- Rodents
- Lion

Type here to search

Question # 9 of 10 (Start time: 02:37:10 PM, 03 March 2022)

Metabolic oxidation of carbohydrates produces _____ as a by-product

Select the correct option

- CO_2
- Water
- O_2
- glucose

Type here to search

ZOO202:Quiz no. 3

Question # 4 of 10 (Start time: 02:47:52 PM, 03 March 2022)

The evolution of similar structures in different lineages is called _____.

Select the correct option

- ○ divergent evolution
- ○ evolution
- ○ similarity of structures
- ○ convergent evolution

Type here to search

BC200204207: SABAHAT

ZOO202:Quiz no. 3

Question # 5 of 10 (Start time: 02:48:52 PM, 03 March 2022)

Elephants belong to which one of the following order?

Select the correct option

- ○ Edentate
- ○ Primates
- ○ Proboscidca
- ○ Lagomorpha

Type here to search

BC200204207: SABAHAT

ZOO202:Quiz no. 3

Question # 8 of 10 (Start time: 02:51:57 PM, 03 March 2022)

Glands develop from the _____ of the skin

Select the correct option

- ○ Epidermis
- ○ Pores
- ○ Hair follicles
- ○ Dermis

Type here to search

quiz.vu.edu.pk/QuizQu

C200202278: QASIM ALI

OO202:Quiz no. 3

uestion # 9 of 10 (Start time: 05:22:46 PM, 03 March 2022)

After a hibernating mammal retreats to a burrow or a nest, the hypothalamus sets the body's thermostat to about ____

elect the correct option

- 4° C
- 2° C
- 8° C
- 6° C

Question # 2 of 10 (Start time: 05:58:28 PM, 03 March 2022)

Cecum commonly observed in _____.

Select the correct option

○	Carnivores
○	Ruminants
○	Human
○	Omnivores

BC190408169: MUBASHIR SATTAR

ZOO202:Quiz no. 3

Question # 7 of 10 (Start time: 05:29:05 PM, 03 March 2022)

The vertebral column of mammals is divided into _____ regions.

Select the correct option

- 5
- 2
- 4
- 3

BC190408169: MUBASHIR SATTAR

Time Left 89 sec(s)

ZOO202:Quiz no. 3

Quiz Start Time: 05:25 PM, 03 March 2022

Question # 5 of 10 (Start time: 05:28:08 PM, 03 March 2022)

Total Marks: 1

The respiratory rate of a hibernating ground squirrel falls from 100 to 200 breaths per minute to about _____ breaths per minute.

Select the correct option

Reload Math Equations

- 8
- 4
- 6
- 2

Click to Save Answer & Move to Next Question

ZOO202:Quiz no. 3

Question # 3 of 10 (Start time: 05:26:53 PM, 03 March 2022)

_____ is a keratinized derivative of the epidermis of the skin.

Select the correct option

- Glands
- Ear
- Hair
- Hormones

BC200204207: SABAHAT

ZOO202:Quiz no. 3

Question # 2 of 10 (Start time: 02:45:56 PM, 03 March 2022)

_____ completely separates the nasal passages from the mouth cavity.

Select the correct option

- (●) Soft palate
- ( ) Pharynx
- ( ) Epiglottis
- ( ) Larynx

Type here to search

quiz.vu.edu.pk/QuizQuestion.aspx?ver=e36f3cd8-cafd-42a8-abd7-

BC200411252: SHAMSA SAEED

ZOO202:Quiz no. 3

Question # 4 of 10 (Start time: 05:32:23 PM, 03 March 2022)

_____ completely separates the nasal passages from the mouth cavity.

Select the correct option

○	Pharynx
●	Soft palate
○	Epiglottis
○	Larynx

Type here to search

BC200411252: SHAMSA SAEED

ZOO202:Quiz no. 3

Question # 9 of 10 (Start time: 05:33:50 PM, 03 March 2022)

Shivering thermogenesis is _____activity.

Select the correct option

- ○ Non-muscular
- ● Muscular
- ○ Simple
- ○ Complex

quiz.vu.edu.pk/QuizQuestion.aspx?ver=c3ca206d-a9e3-41f9-96e8-58fd8ef4e101

BC200201950: MUHAMMAD HABIB

ZOO202:Quiz no. 3

Question # 5 of 10 (Start time: 05:41:46 PM, 03 March 2022)

_____ is a penod of winter inactivity in which the hypothalamus of the brain slows the metabolic, heart, and respiratory rates

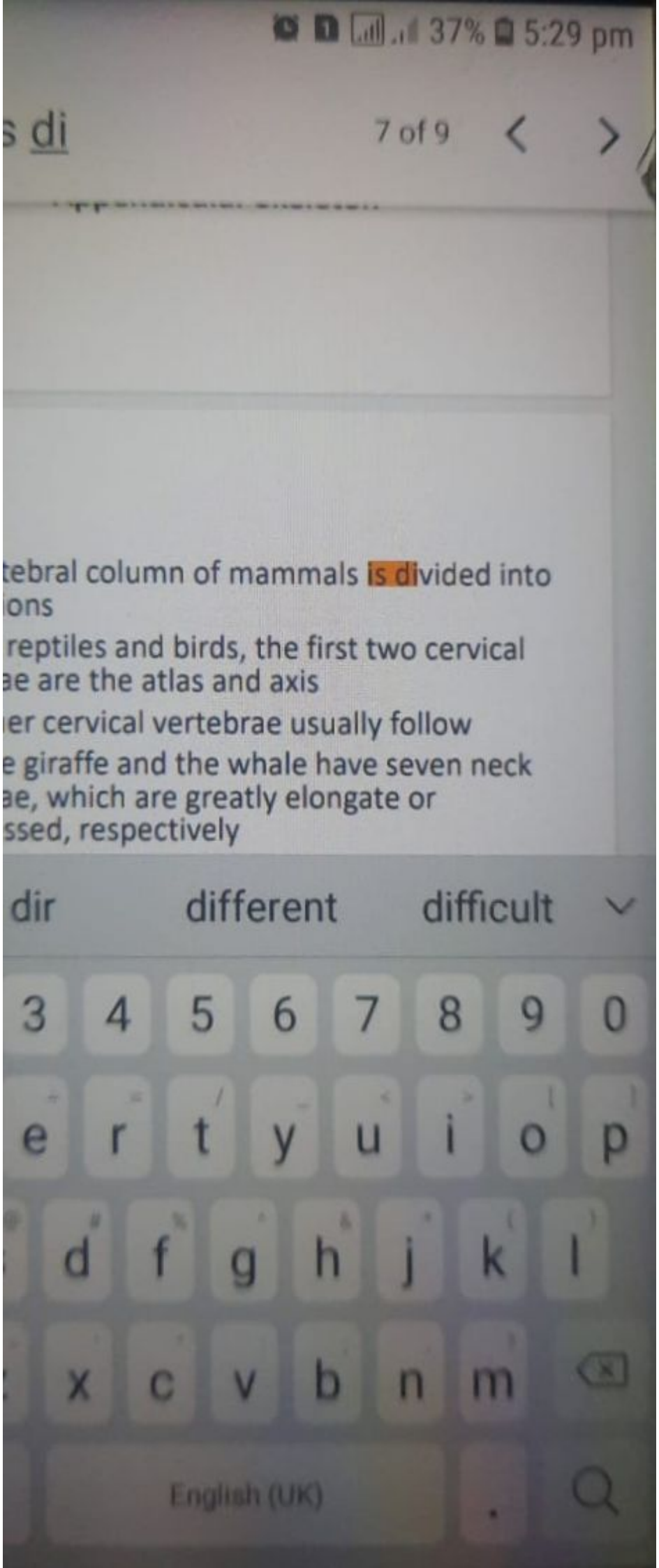
Select the correct option

- ○ Aestivation
- ○ Winter sleep
- ● Hibernation
- ○ Summer sleep

Type here to search

Digestive System

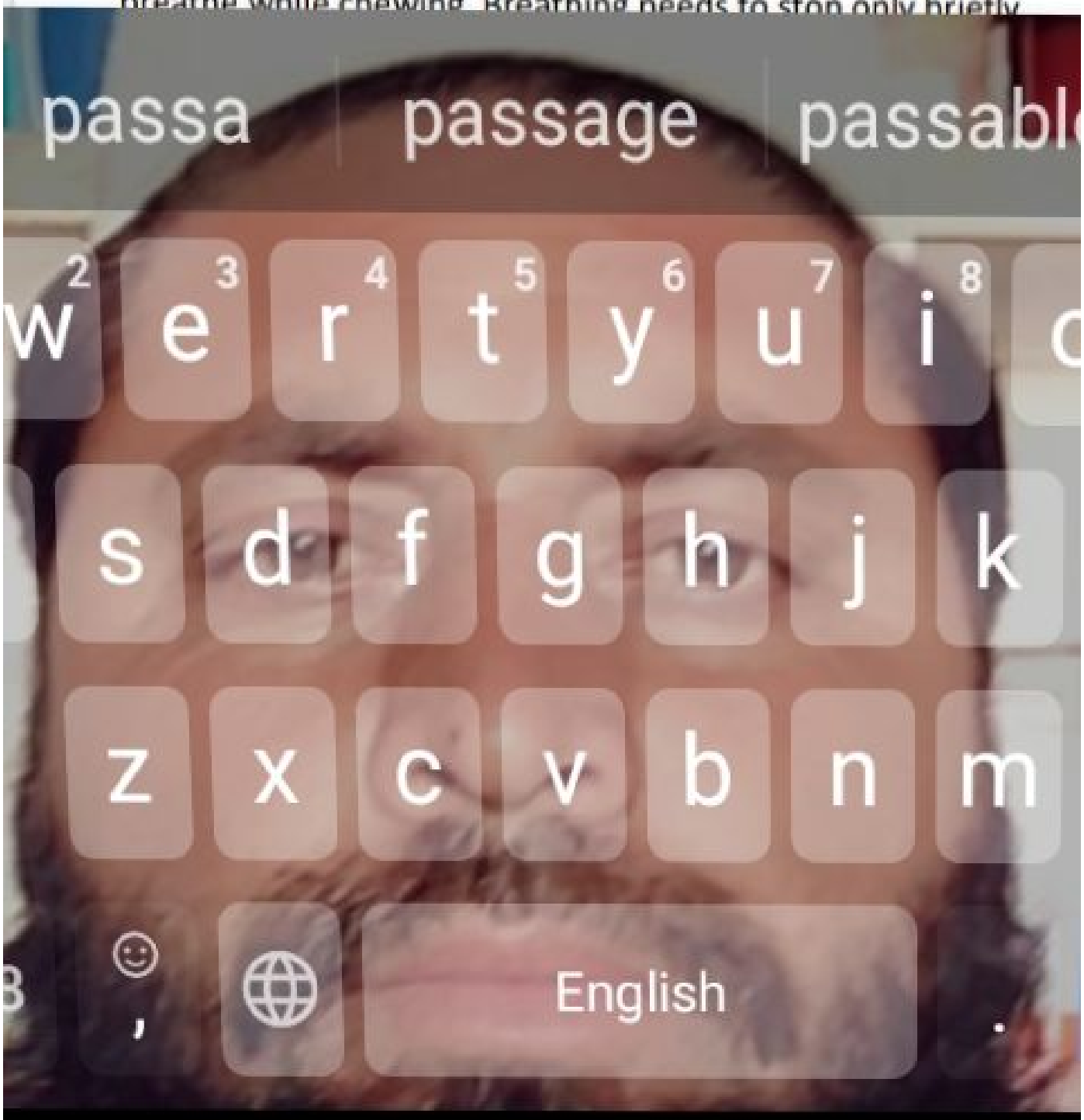
- Specializations in the digestive tract of most herbivores reflect the difficulty of digesting food rich in cellulose
- Horses, rabbits, and many rodents have an enlarged **cecum** at the junction of the large and small intestines
- A **cecum** is a fermentation pouch where microorganisms aid in cellulose digestion. Sheep, cattle, and deer are called **ruminants** (L. ruminare, to chew the cud). Their stomachs are modified into four chambers

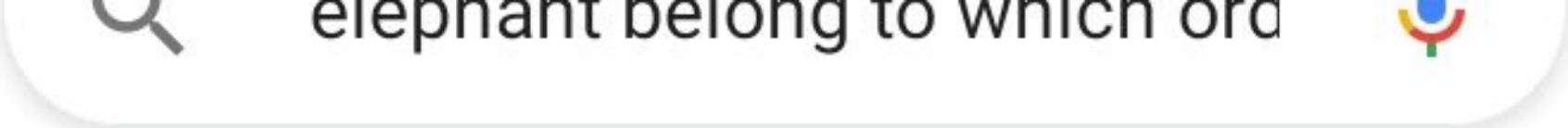

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7 of 9
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English (UK)

Hibernation

- **Hibernation** is a period of winter inactivity in which the hypothalamus of the brain slows the metabolic, heart, and respiratory rates
- True hibernators include the monotremes (echidna and duck-billed platypus) and many members of the Insectivora (e.g., moles and shrews), Rodentia (e.g., chipmunks and woodchucks), and Chiroptera (bats)
- In preparation for hibernation, mammals usually accumulate large quantities of body fat.

- A **secondary palate** evolved twice in vertebrates—in the archosaur lineage and in the synapsid lineage
- In some therapsids, small, shelflike extensions of bone (the hard palate) partially separated the nasal and oral passageways
- In mammals, the secondary palate extends posteriorly by a fold of skin, called the **soft palate**, which almost completely separates the nasal passages from the mouth cavity. Unlike other vertebrates that swallow food whole or in small pieces, some mammals chew their food
- The more extensive secondary palate allows mammals to breathe while chewing. Breathing needs to stop only briefly





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Order Proboscidea

Elephants are the only living members of the **Order Proboscidea** today.

https://seaworld.org › animals › clas...

All About Elephants - Scientific Classification - SeaWorld.org

About featured snippets Feedback

- Condensation as warm air in the respiratory passages encounters the cooler nasal passages minimizes respiratory water loss
- A low-protein diet, which reduces urea production, minimizes excretory water loss
- The nearly dry seeds that the kangaroo rat eats are rich sources of carbohydrates and fats
- Metabolic oxidation of carbohydrates produces water as a by-product.

- The hearts of birds and mammals are superficially similar
- Both are four-chambered pumps that keep blood in the systemic and pulmonary circuits separate, and both evolved from the hearts of ancient reptiles
- Their similarities, however, are a result of adaptations to active lifestyles
- The evolution of similar structures in different lineages is called **convergent evolution**

- The mammalian heart evolved in the synapsid

- Claws are present in all amniote classes. They are used for locomotion and offensive and defensive behavior. Claws form from accumulations of keratin that cover the terminal phalanx (bone) of the digits. In some mammals, they are specialized to form nails or hooves (figure 22.7).
- **Glands** develop from the epidermis of the skin
- **Sebaceous (oil) glands** are associated with hair follicles, and their oily secretion lubricates and waterproofs the skin and hair



- The transport processes in this loop and the remainder of the tubule system allow mammals to produce urine that is more concentrated than blood
- For example, **beavers** produce urine that is **twice** as concentrated as blood, while **Australian hopping mice** produce urine that is **22 times** more concentrated than blood
- This accomplishes the same function that nasal and orbital salt glands do in reptiles and birds.

Temperature regulation

- Mammals are widely distributed over the earth, and some face harsh environmental temperatures
- Nearly all face temperatures that require them to dissipate excess heat at some times and to conserve and generate heat at other times
- Heat-producing mechanisms of mammals are divided into two categories
- **Shivering thermogenesis** is muscular activity that generates large amounts of heat but little movement

- **Nonshivering thermogenesis** involves heat production by general cellular metabolism and the metabolism of special fat deposits called brown fat
- Heat production is effective in thermoregulation because mammals are insulated by their pelage and/or fat deposits
- Fat deposits are also sources of energy to

Hibernation

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- In preparation for hibernation, mammals usually accumulate large quantities of body fat.

- After a hibernating mammal retreats to a burrow or a nest, the hypothalamus sets the body's thermostat to about 2° C
- The respiratory rate of a hibernating ground squirrel falls from 100 to 200 breaths per minute to about four breaths per minute
- The heart rate falls from 200 to 300 beats per minute to about 20 beats per minute
- During hibernation, a mammal may lose a third to half of its body weight

- Mammals, like all amniotes, have a metanephric kidney
- Unlike reptiles and birds, which excrete mainly uric acid, mammals excrete urea
- Urea is less toxic than ammonia and does not require large quantities of water in its excretion
- Unlike uric acid, however, urea is highly water soluble and cannot be excreted in a semisolid form; thus, some water is lost
- Excretion in mammals is always a major route for water loss.

- Auditory senses were similarly important to early mammals
- More recent adaptations include an ear flap (the pinna) and the external ear canal leading to the **tympanum** that directs sound to the middle ear
- The middle ear contains three ear ossicles that conduct vibrations to the inner ear
- The sensory patch of the inner ear that contains the sound receptors is long and coiled and is called the **cochlea**
- This structure provides more surface area for receptor cells and gives mammals greater sensitivity to pitch and volume than is present in reptiles
- Cranial casts of early mammals show well-developed auditory regions.

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